

**Modeling of High Risk
Indicators of Certification
Error in the National School
Lunch Program**

Draft Variable List

October 2, 2009

Nancy Cole
Quinn Moore



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U.S. Department of Agriculture
Food and Nutrition Service
3101 Park Center Drive
Alexandria, VA 22302
Project Officer: Sheku Kamara

Submitted by:
Mathematica Policy Research
955 Massachusetts Avenue
Suite 801
Cambridge, MA 02139
Telephone: (617) 491-7900
Facsimile: (617) 491-8044
Project Director: Nancy Cole

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MODELING OF HIGH RISK INDICATORS OF CERTIFICATION ERROR

The Food and Nutrition Service (FNS), U.S. Department of Agriculture, has contracted with Mathematica Policy Research to identify local education authorities or agencies (LEAs) at high risk for certification error, describe the characteristics of error prone LEAs, model the factors associated with high risk for certification error, and develop a monitoring tool for States to use in identifying LEAs at high risk.

This deliverable provides a variable list for the analytic files that we will compile for the study. After receiving comments from FNS, the variable list will be incorporated in our revised study design. A graphic presentation of the analytic files planned for the study is shown in Figure 1. These files will be used for two purposes:

1. ***Develop a Model of High-Risk Indicators of Certification Error for Public LEAs.***

These models will be based on existing secondary data. We will use multiple data sources—Access, Participation, Eligibility, and Certification Study (APEC) data, Regional Office Review of Applications (RORA) data, and Verification Summary Reports (VSR)—to examine different types of certification error (household reporting error, administrative error, and overall error). Multiple years of VSR and RORA data will be used to examine changes in factors associated with certification error over time.

2. ***Develop a Monitoring Tool for Use by State Agencies.*** This software tool will be used to apply model parameters and automate the process of annually evaluating LEAs relative to multiple risk criteria. State agencies will use the tool to evaluate current-year VSR data and National School Lunch Program (NSLP) reimbursements, and output a ranking of LEAs by risk for certification error. The tool will enable States to sort or filter the listing of “high-risk” LEAs by NSLP reimbursement or geographic area, or other factors, so that they can readily identify LEAs that satisfy FNS criteria for “Additional Administrative Review” (SP 01-2009) and to select LEAs that can be reviewed within the resources available in their travel budgets.

A. Sources of Data for Analytic Files

Table 1 provides a list of the eight data sources that will be used to compile analytic files for the study. Six of these data sources include information about the universe of public school districts: (1) Verification Summary Reports (VSR), (2) Common Core of Data (CCD) LEA Universe Survey, (3) CCD LEA Finance Survey, (4) Census Small Area Income and Poverty Estimates, (5) Bureau of Labor (BLS) Local Area Unemployment Survey, and (6) FNS National Data File. APEC and RORA data, which provide key measures of certification error, include samples of public school districts.

The analytic files will include annual cross sections for school years 2004-2005 through 2008-2009. APEC data are available for a single school year (2005-2006) while all other data files are available for multiple years.¹

¹ We request clarification from FNS regarding the availability of SY2008-2009 VSR and RORA data.

B. List of Variables for Analytic Files

The variable list is presented in two tables:

- Table 2 provides an overview of the categories of measures (for example, measures of certification error), the types of measures within category (e.g., household reporting error, administrative error, and overall error), the justification for including the measure in the analytic file, and the sources of data.
- Table 3 is organized by data source and measure domain within data source. This table provides a description of detailed measures, the corresponding variable names, whether the measure will be used in its raw form or as input to constructed measures, and the measure category. The school year for which measures are available is indicated after the name of the data source.

The variable list includes measures of certification error to be used as dependent variables in the estimation models, and LEA characteristics that will be considered for inclusion as covariates in the models. Our modeling effort will combine theory-based and automated approaches to variable selection and model development. Table 3 does not provide an indication of the variables to be included in the monitoring tool because development of the monitoring tool begins after Mathematica and FNS have agreed on final model specifications. All variables from the final models will be incorporated in the monitoring tool.

Figure 1. Overview of Study Approach

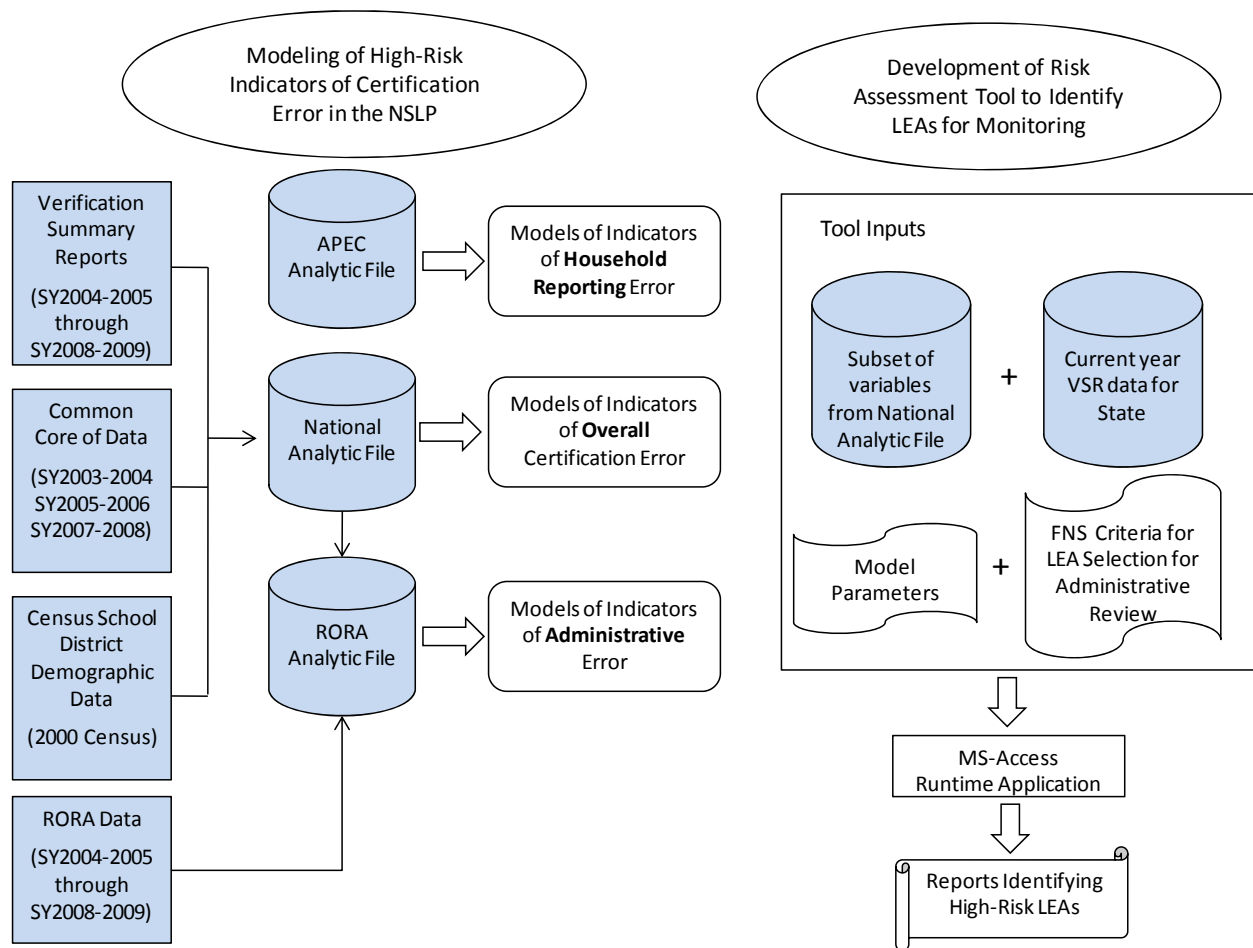


Table 1. Data Sources for Analytic Files

School Year	Verification Summary Reports	RORA Application Data	Application, Eligibility, and Certification Study Data (APEC)	Food and Nutrition Service National Data File	Common Core of Data, LEA Universe Survey	Common Core of Data, LEA Finance Survey	Small Area Income and Poverty Estimates	Local Area Unemployment Survey
SY2004-2005	2004-2005	2005	SY2005-2006	FY2004	2004-2005	FY2004	Annual data for counties and school districts	Annual data for counties
SY2005-2006	2005-2006	2006		FY2005	2005-2006	FY2005		
SY2006-2007	2006-2007	2007		FY2006*	2006-2007	FY2006		
SY2007-2008	2007-2008	2008*		FY2007*	NA	FY2007		
SY2008-2009	2008-2009*	2009*		FY2008	Preliminary file available	Not yet available		

*Indicates that data have not been received from FNS.

Note: Years are shown as school year, fiscal year, or year corresponding to file naming conventions for each data source.

Table 2. List of Variables, Justification for Inclusion, and Data Sources

Variable	Justification	Data Sources
Measures of Certification Error		
Household reporting error	Necessary for answering questions related to reporting error.	APEC
Administrative error	Necessary for answering questions related to administrative error.	APEC, RORA
Overall certification error	Necessary for answering questions related to overall certification error.	APEC, VSR
Verification Results		
Verification sampling method	LEAs using random sampling may have different certification rates than those using error-prone sampling.	VSR
Verification results by certification category	Previous research has shown verification results to be highly related to certification error.	VSR
LEA Meal Program Characteristics		
Use of direct certification and type of direct certification used	Students directly certified for benefits are less likely to be subject to certification error because no household reporting or application processing is involved.	APEC
Number of students not subject to verification	This measure is closely related to direct certification.	VSR
Number of students categorically eligible for benefits	Applications based on categorical eligibility may have lower rates of administrative error because less processing is involved.	VSR
Use of Provisions 2 or 3	Schools using Provisions 2 or 3 and in their base year have strong incentives to increase the number of students certified for meal benefits.	VSR, APEC
Percentage of students certified for free or reduced-price meals	LEAs with more applications for meal benefits may have higher levels of certification error.	VSR, CCD
Number of meals by benefit type	Enables calculation of dollar amounts associated with certification error rates.	APEC, FNSND, APECSA

Table 2 (continued)

Variable	Justification	Data Sources
LEA Demographic Characteristics		
Enrollment and number of schools	Certification error may vary by district size.	VSR, CCD
Percentage of Title I-eligible schools	Districts with more Title I-eligible schools likely have more disadvantaged students, and thus more students eligible for meal benefits.	CCD
Grade distribution and span	Certification error may differ according to whether districts have older or younger students.	CCD
Racial/Ethnic Composition	Race and ethnicity are factors highly associated with a number of important outcomes.	CCD
LEA Administrative Characteristics		
Number of SFA administrators and support staff	Districts with more administrators and support staff may be more able to process school meal benefit applications effectively.	CCD
Spending on food services, food service salaries, and administrative support services	These variables may proxy for the amount of resources devoted to food services, which may be related to their effectiveness in processing applications.	CCD
Local Characteristics		
Region	Different regions may exhibit higher or lower levels of certification error.	CCD
Urbanicity	LEA's may exhibit different levels of certification error depending on whether they are located in more urban or more rural areas.	CCD
County school-age poverty rate and median household income	These variables may be related to the number of students eligible for school meal benefits and to the resources available to the LEA.	SAIPE
County unemployment rate	LEAs in areas with higher unemployment rates may experience higher error rates	LAUS

APEC = Application, Eligibility, and Certification; APECSA = APEC State agency reports; CCD = Common Core of Data; FNSND = Food and Nutrition Service National Datafile; LAUS = Local area unemployment statistics (Bureau of Labor Statistics); RORA = Regional Office Review of Applications; SAIPE = Small Area Income and Poverty Estimates (Census Bureau); SFA = school foodservice authority; VSR = Verification Summary Report.

Table 3. List of Measures to Include in Analytic Files for Consideration in Models of High Risk Indicators of Certification Error in the NSLP

Domain / Measure	Raw Variable Names	Use Raw Measure?	Constructed Measures Based on Raw Measure	Purpose
Application, Participation, Eligibility, and Certification Study Data, SY2005-2006 ^a				
Measures of Overall Certification Error^b				
Pct. free meals served to reduced price eligible students	cfrepct_l	Yes	n.a.	Measure of certification error
Pct. free meals served to students not eligible for benefits	cfpepct_l	Yes	n.a.	Measure of certification error
Pct. reduced price meals served to free eligible students	crfepct_l	Yes	n.a.	Measure of certification error
Pct. reduced price meals served to students not eligible for benefits	crpepct_l	Yes	n.a.	Measure of certification error
Pct. paid meals served to free eligible students	cpfepct_l	Yes	n.a.	Measure of certification error
Pct. paid meals served to reduced price eligible students	cprepct_l	Yes	n.a.	Measure of certification error
Measures of Reporting Error				
Pct. free meals served to reduced price eligible students	cfrepct_rpt_l	Yes	n.a.	Measure of certification error
Pct. free meals served to students not eligible for benefits	cfpepct_rpt_l	Yes	n.a.	Measure of certification error
Pct. reduced price meals served to free eligible students	crfepct_rpt_l	Yes	n.a.	Measure of certification error
Pct. reduced price meals served to students not eligible for benefits	crpepct_rpt_l	Yes	n.a.	Measure of certification error
Pct. paid meals served to free eligible students	cpfepct_rpt_l	Yes	n.a.	Measure of certification error
Pct. paid meals served to reduced price eligible students	cprepct_rpt_l	Yes	n.a.	Measure of certification error

Table 3 (continued)

Domain / Measure	Raw Variable Names	Use Raw Measure?	Constructed Measures Based on Raw Measure	Purpose
Measures of Administrative Error				
Pct. free meals served to reduced price eligible students	cfrepct_adm_l	Yes	n.a.	Measure of certification error
Pct. free meals served to students not eligible for benefits	cfpepct_adm_l	Yes	n.a.	Measure of certification error
Pct. reduced price meals served to free eligible students	crfepct_adm_l	Yes	n.a.	Measure of certification error
Pct. reduced price meals served to students not eligible for benefits	crpepct_adm_l	Yes	n.a.	Measure of certification error
Pct. paid meals served to free eligible students	cpfepct_adm_l	Yes	n.a.	Measure of certification error
Pct. paid meals served to reduced price eligible students	cprepct_adm_l	Yes	n.a.	Measure of certification error
LEA Characteristics				
LEA uses direct certification	sfa_direct_cert	Yes	n.a.	Covariate in models
Method for direct certification (no match, district-level match, state-level match, other)	sfa_dcert_method	Yes	n.a.	Covariate in models
LEA uses computer software to process and determine eligibility	sfa_comp_proc	Yes	n.a.	Covariate in models
LEA uses food management company	sfa_foodmanage	Yes	n.a.	Covariate in models
LEA tracks individual student participation	sfa_part_data	Yes	n.a.	Covariate in models
Verification Summary Report, SY2004-2005 through SY2008-2009				
Verification-related variables				
Method of selecting verification sample	typeofverif	Yes	n.a.	Covariate in models, and used as interaction term with verification results
Free, categorically eligible applications verified	a1, a3, a5, a7	No	Percentage of sampled applications in this category	Covariate in models and input in variable construction

Table 3 (continued)

Domain / Measure	Raw Variable Names	Use Raw Measure?	Constructed Measures Based on Raw Measure	Purpose
Free, income-eligible applications verified	b1,b3, b5, b7	No	Percentage of sampled applications in this category	Covariate in models and input in variable construction
Reduced price applications verified	c1, c3, c5, c7	No	Percentage of sampled applications in this category	Covariate in models and input in variable construction
Applications not responding to verification	a7, b7, c7	No	Percentage of sampled applications in this category	Covariate in models and input in variable construction
Verification results	a1, a3, a5, a7, b1,b3, b5, b7, c1, c3, c5, c7	No	Percentage or number of sampled applications in error, overall, by approval category, and by result category	Measures of certification error and covariates in models
LEA Characteristics				
Number of schools	Numnslpschools	Yes	n.a.	Covariate in models
Number of non-base year P23 schools	Numprovisionschools	Yes	Percentage of schools	Covariate in models
Enrollment	Enrollment	Yes	n.a.	Covariate in models
Students enrolled in non-base year P23 schools	enrollmentprovision	Yes	Percentage of enrolled students	Covariate in models
Meal benefit certification composition	freeligtot, rpelig, enrollment	Yes	Percentage of enrolled students	Covariate in models
Free-certified students not subject to verification	freelignotverified	Yes	Percentage of enrolled students and of free-certified students	Covariate in models
Categorically eligible free-certified students	Freeeligcat	Yes	Percentage of enrolled students and of free-certified students	Covariate in models
Free-certified students for non-base year P23 schools	freereportedprov	Yes	Percentage of enrolled students and of free-certified students	Covariate in models

Table 3 (continued)

Domain / Measure	Raw Variable Names	Use Raw Measure?	Constructed Measures Based on Raw Measure	Purpose
Reduced price-certified students for non-base year P23 schools	rpreortedprov	Yes	Percentage of enrolled students and of reduced price-certified students	Covariate in models
Number of approved applications	freecateligapps, freeincomeapps, rpapps	Yes	n.a.	Covariate in models
Categorically eligible applications	freecateligapps	Yes	Percentage of enrolled students and of free-certified applications	Covariate in models
Applications approved for free meals based on income	freeincomeapps	Yes	Percentage of enrolled students and of free-certified applications	Covariate in models
Applications approved for reduced price meals	Rpapps	Yes	Percentage of enrolled students and of reduced price-certified applications	Covariate in models
Regional Office Review of Applications, SY2004-2005 through SY2008-2009^a				
Measures of Administrative Error				
For each sampled application, indicator of approval category, and SFA and FNS eligibility determination	catelig, sfaelig, fnselig	No	Percent of free-approved students certified in error	Measure of certification error
For each sampled application, indicator of approval category, and SFA and FNS eligibility determination	catelig, sfaelig, fnselig	No	Percent of RP-approved students certified in error	Measure of certification error
LEA Administrative Characteristics (not available for SY2004-2005)				
Applications stored centrally or at each school	ApplicationStorage	Yes	n.a.	Covariate in models
Application type: multi-child or individual	ApplicationType	Yes	n.a.	Covariate in models

Table 3 (continued)

Domain / Measure	Raw Variable Names	Use Raw Measure?	Constructed Measures Based on Raw Measure	Purpose
Common Core of Data, SY2004-2005 through SY2008-2009 ^c				
LEA Demographic Characteristics				
Enrollment	memberYR	Yes	Number of students per school.	Covariate in models
Race/ethnicity composition	asianYR, hispYR, blackYR, whiteYR	No	Percentage of enrolled students in each category.	Covariate in models
Migrant student composition	migrantYR	No	Percentage of enrolled students.	Covariate in models
Percentage of students eligible for free or reduced price meals	frelchYR, redlchYR	Yes	n.a.	Covariate in models
Urbanicity	localeYR	No	Raw variable includes 8 categories. Constructed variable will aggregate to fewer categories	Covariate in models
LEA Administrative Characteristics				
Number of schools	schYR	Yes	n.a.	Covariate in models
Number of teachers	tottchYR	Yes	n.a.	Covariate in models
Grade distribution and span	pkYR, kgYR, g01YR, g02YR, g03YR, g04YR, g05YR, g06YR, g07YR, g08YR, g09YR, g10YR, g11YR, g12YR, gshiYR, gsloYR	No	Percentage of students in various grade ranges	Covariate in models
Number of LEA administrators	leadmYR	Yes	Number per student.	Covariate in models
Number of school administrators	schadmYR	Yes	Number per student.	Covariate in models
Number of school support staff	schsupYR	Yes	Number per student.	Covariate in models
Food services spending	e11YR	Yes	Spending per student.	Covariate in models
Food service salary spending	v29YR	Yes	Spending per student.	Covariate in models

Table 3 (continued)

Domain / Measure	Raw Variable Names	Use Raw Measure?	Constructed Measures Based on Raw Measure	Purpose
Administrative support services spending	v15YR	Yes	Spending per student.	Covariate in models
School administrative support services spending	v17YR	Yes	Spending per student.	Covariate in models
Census Bureau Small Area Income and Poverty Estimates, SY2004-2005 through SY2008-2009				
Area Characteristics				
County File: Overall county poverty rate	Estimated percent of people of all ages in poverty	Yes	n.a.	Covariate in models
School District File: LEA school-age poverty rate	Population age 5 – 17, Estimated number of children 5 to 17 years old in poverty	No	Percentage of children age 5-17 years old in poverty	Covariate in models
Bureau of Labor Statistics Local Area Unemployment Survey, SY2004-2005 through SY2008-2009				
County unemployment rate	Unemployment rate	Yes	n.a.	Covariate in models
Food and Nutrition Service National Data File, SY2004-2005 through SY2008-2009				
State-level numbers of meals by benefit type	NSLP free lunches, NSLP reduced price lunches, NSLP paid lunches	No	Imputed number of LEA meals by benefit type	Input in variable construction

^aAdditional information on the LEAs in this dataset will be drawn from other sources.

^bMore broadly defined certification error measures, such as the percentage of all meals in error due to overall, reporting, or administrative error, will be created based on the more finely defined measures listed here.

^cIncludes items from LEA universe survey, LEA finance survey, school universe survey. Items from the school universe survey will be summed to the LEA level. The “YR” suffix on the raw variables listed in this section refer to the year of the CCD data from which the variables are drawn.

NSLP=National School Lunch Program; n.a=Not applicable; P23=Provisions 2 or 3.

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